

The Hong Kong Polytechnic University
Department of Electronic and Information Engineering

EIE3320 Homework 7: Java Web Programming (Servlets and JSP)

(Deadline for Submission: Check the course information)

Table of Contents

1. Preparation for Homework 7 and Lab 2 (with backup and restore MySQL database)
2. Homework 7
3. APPENDIX 1 - Troubleshooting in EEE Laboratories
4. APPENDIX 2 - MySQL Commands
5. APPENDIX 3 - Common SQL Statements
6. APPENDIX 4 – Special setting to connect MySQL database pre-installed in CF502/CF504
7. APPENDIX 5 – Special setting to change MySQL port number

Section 1 – Preparation for Homework 7 and Lab 2

Introduction

The following procedures assume that you do not have the administrative right but you can write to C:\temp. Remember to compress the folder C:\Temp\EIE3320 and copy the compressed file to your flash memory before leaving the lab. The procedures have been tested under Windows 11. Tomcat and Eclipse will also work on Mac OS and Linux, but you have to download the programs by yourself.

Software requirements

Eclipse IDE for Enterprise Java and Web Developers 2024-12, Tomcat 11.0.4, MySQL Community Server 8.4.3, MySQL Connector/J 8.4.0

(Note: In the EEE computer laboratory, Eclipse has been installed, and all the files mentioned above have been saved in the C:\Temp folder that you do not need to download again. If you work at home, you can download them from Blackboard.)

Downloading and Installing Software

1. Create a folder named “C:\Temp\EIE3320”
Create a folder named “C:\Temp\EIE3320\eclipse-workspace” under your home folder in CD514
Create a folder named “C:\Temp\EIE3320\MySQL”
2. Download Apache Tomcat 11 from Blackboard and unzip the file to C:\Temp\EIE3320.

3. Download Eclipse zip file and unzip the file to C:\Temp\EIE3320. (Skip this step if Eclipse has been installed.)
4. Download mysql-8.4.3-winx64.zip from Blackboard and unzip it to C:\Temp\EIE3320\MySQL.
5. Download MySQL Connector (mysql-connector-j-8.4.0.zip) from Blackboard and unzip the file to C:\Temp\EIE3320\MySQL.
6. Download a SQL script (Books.sql) from Blackboard and copy the file to C:\temp\EIE3320\MySQL.
7. Add the MySQL JDBC driver to Tomcat by **copying “C:\Temp\EIE3320\MySQL\mysql-connector-j-8.4.0\mysql-connector-j-8.4.0.jar” to “C:\Temp\EIE3320\apache-tomcat-11.0.4\lib”**. This .jar file contains classes that allow Java programs to manipulate MySQL databases. A JAR file (Java Archive File) is a compressed file that contains a number of .class files in a package.

Setting up MySQL Accounts

8. Open a Command Window (Press the **Windows** key + **r**, type *cmd*) and execute the following command:

cd C:\Temp\EIE3320\MySQL\mysql-8.4.3-winx64\bin

```
C:\>cd C:\Temp\EIE3320\MySQL\mysql-8.4.3-winx64\bin
C:\Temp\EIE3320\MySQL\mysql-8.4.3-winx64\bin>
```

*The *cd* command, known as Change Directory, can be used to change the current working directory in various operating systems. Several use cases:

Commands	Description
<i>cd <child directory></i>	Go to child directory
<i>cd..</i>	Back to parent directory
<i>cd\</i>	Back to root directory

*To switch between different drives, e.g. C drive to P drive, you may type *P:* to change the drive directory. To change back to C drive, you can type *C:*

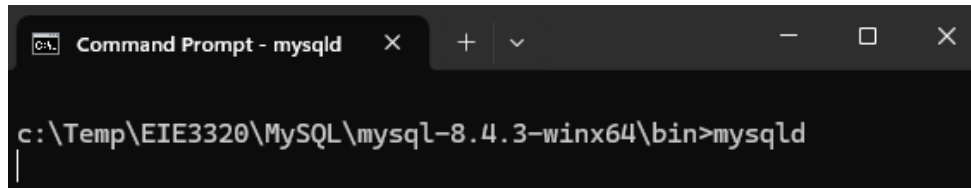
Then, execute the following command in the Command Window to start the MySQL server:

mysqld --initialize-insecure

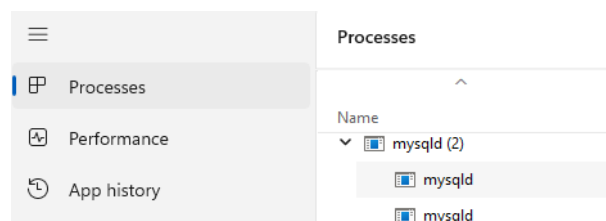
```
C:\Temp\EIE3320\MySQL\mysql-8.4.3-winx64\bin>mysqld --initialize-insecure
```

*Two dashes "--" before "initialize", this command will create a data folder in mysql-8.4.3-winx64 folder. Running the command may take a minute, please wait until it is finished. If the command window is freezing for a long time or crashed, try rebooting the command window or your PC. Then, type the command:

mysqld



The above prompt indicates that the MySQL server is running. Keep the command window open to maintain the server's operation. You won't be able to enter additional commands or reuse this command prompt. To verify the process, open the Task Manager by right-clicking the taskbar at the bottom of the screen and selecting "Task Manager." Then, go to the "Processes" tab to check if mysqld.exe is running.



9. After the MySQL server is started, open another new Command Window. Change directory to MySQL and start the MySQL monitor.

cd C:\Temp\EIE3320\MySQL\mysql-8.4.3-winx64\bin

mysql -h localhost -u root



“root” is the default user account that is created during installation. The --h option is used to specify the host that runs mysqld.exe.

10. At the mysql> prompt, type

USE mysql

Note that mysql is a build-in database.

11. Add the user “guest” to the mysql build-in database. You may copy-and-paste the following command to your Command Windows running mysql.exe.

```
create user 'guest'@'localhost' identified by 'guest';  
grant select, insert, update, delete, create, drop, references,  
execute on *.* to 'guest'@'localhost';  
exit;
```

12. Shutdown mysqld.

```
mysqladmin -h localhost -u root shutdown
```

You will notice that mysqld.exe running in another command window is terminated.

Creating a Book Database

13. Restart the MySQL server.

```
cd C:\Temp\EIE3320\MySQL\mysql-8.4.3-winx64\bin  
mysqld --initialize-insecure  
mysqld
```

Then, open another command window to start the MySQL monitor (client) by typing

```
cd C:\Temp\EIE3320\MySQL\mysql-8.4.3-winx64\bin  
mysql -h localhost -u guest -p
```

When prompt for entering password, type

```
guest
```

14. Create the Book database by typing

```
source C:/Temp/EIE3320/Books.sql;
```

Note that the semicolon is very important. This command will create a database called “Books” in C:\temp\EIE3320\MySQL\mysql-8.4.3-winx64\data\books. Note also the use of forward slash ‘/’ for separating directories and filenames in a path. You can use Notepad to read C:\temp\EIE3320\ Books.sql.

15. Make sure that the database has been successfully created by typing

```
USE books  
select * from authors;
```

You should see the following table:

```
mysql> select * from authors;
+-----+-----+-----+
| AuthorID | FirstName | LastName |
+-----+-----+-----+
| 1 | Harvey | Deitel |
| 2 | Paul | Deitel |
| 3 | Andrew | Goldberg |
| 4 | David | Coffnes |
+-----+-----+-----+
4 rows in set (0.00 sec)
```

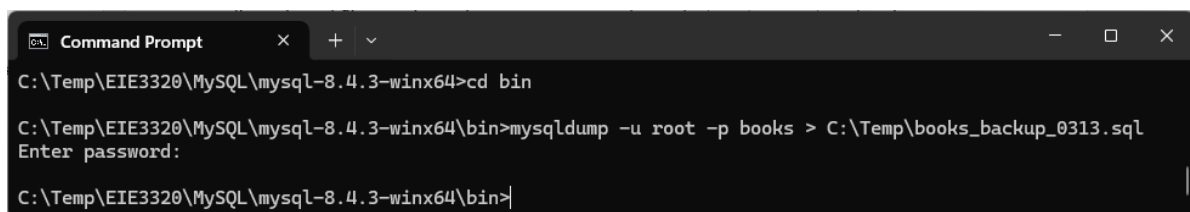
16. Exit MySQL by typing

exit;

(If you wish to make query in a short time, keep this windows)

Generate the backup of a single database

To generate a backup of a single database, run the following command in a new Command Prompt. This command will create a backup of the "books" database, including its structure and data, in the books_backup_0313.sql file. If there is no password set in MySQL, simply press Enter again.



```
Command Prompt
C:\Temp\EIE3320\MySQL\mysql-8.4.3-winx64>cd bin
C:\Temp\EIE3320\MySQL\mysql-8.4.3-winx64\bin>mysqldump -u root -p books > C:\Temp\books_backup_0313.sql
Enter password:
C:\Temp\EIE3320\MySQL\mysql-8.4.3-winx64\bin>
```

```
mysqldump -u root -p books > C:\Temp\books_backup_0313.sql
```

(You can use Notepad to read the content of the file)

Restore the MySQL Database (only if your database has been corrupted or dropped)

In Command Windows running mysql.exe,

```
mysql> show databases;
+-----+
| Database |
+-----+
| books |
| information_schema |
| mysql |
| performance_schema |
| sys |
+-----+
5 rows in set (0.00 sec)
```

```
mysql> CREATE DATABASE IF NOT EXISTS books;
```

```
Query OK, 1 row affected, 1 warning (0.01 sec)
```

```
mysql> Use books;
```

```
Database changed
```

```
mysql> SOURCE C:\Temp\books_backup_0313.sql;
```

```
Query OK, 0 rows affected (0.00 sec)
```

```
Query OK, 0 rows affected (0.00 sec)
```

```
...
```

```
Query OK, 0 rows affected (0.00 sec)
```

```
mysql> show tables;
```

```
+-----+
```

```
| Tables_in_books |
```

```
+-----+
```

```
| authorisbn      |
```

```
| authors         |
```

```
| bookinfo        |
```

```
+-----+
```

```
3 rows in set (0.00 sec)
```

```
mysql> select * from authors;
```

```
+-----+-----+-----+
```

```
| AuthorID | FirstName | LastName |
```

```
+-----+-----+-----+
```

```
|          1 | Harvey    | Deitel   |
```

```
|          2 | Paul      | Deitel   |
```

```
|          3 | Andrew    | Goldberg |
```

```
|          4 | David     | Choffnes |
```

```
+-----+-----+-----+
```

```
4 rows in set (0.00 sec)
```

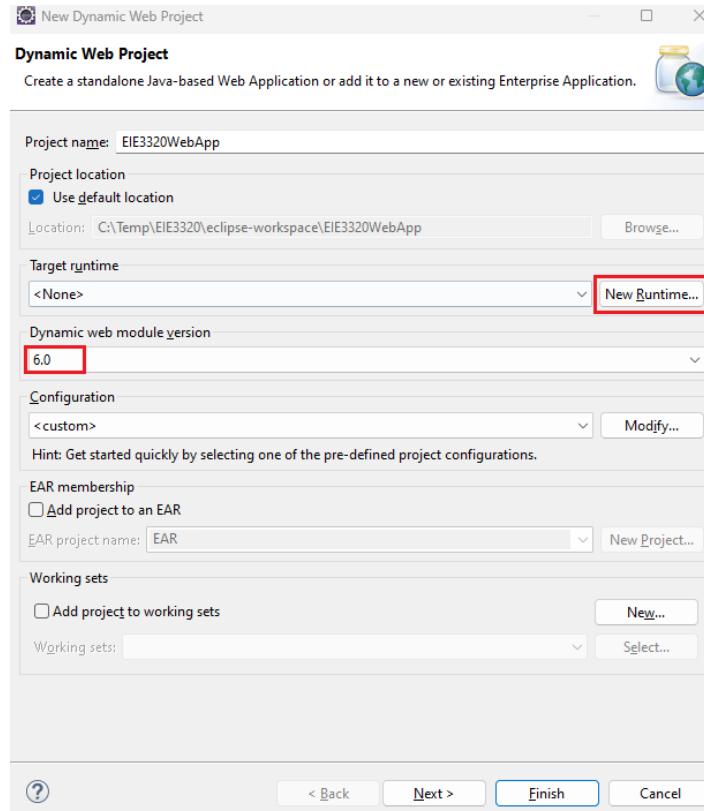
```
mysql>
```

Setting up Eclipse

17. Start Eclipse through the shortcut (**Lab PC Shortcut: Desktop/Apps/Eclipse 2024-12**). If you use the standalone Eclipse, you can start Eclipse by clicking eclipse.exe in C:\Temp\EIE3320\eclipse. If it is the first time you start eclipse, it will prompt you to select a workspace. You may select C:\Temp\EIE3320\eclipse-workspace as the workspace. When you see Eclipse's welcome page, click "Hide" in the top-right corner. You may also like to go through the tutorials (if you have time).

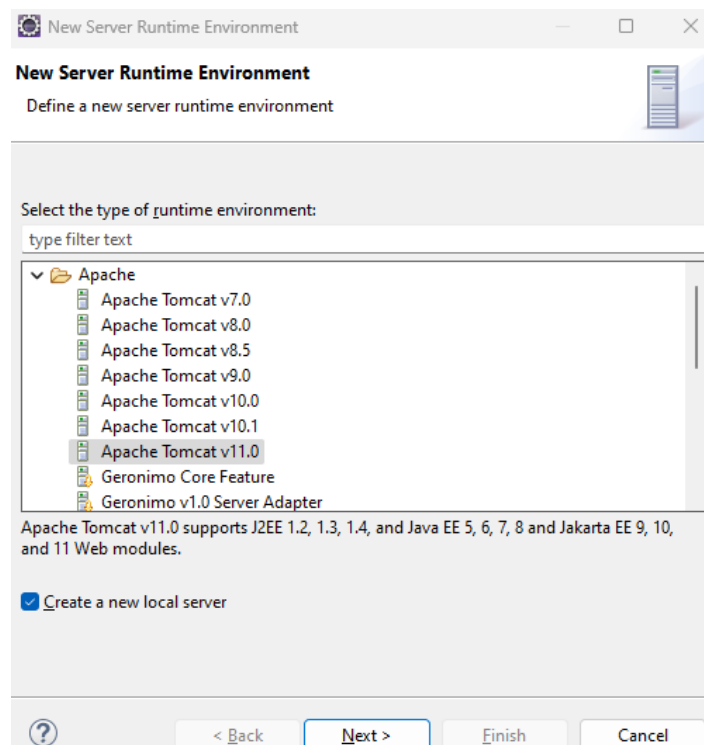
Create Web Project in Eclipse

18. Create a new Dynamic Web project called "EIE3320WebApp" using File->New->Dynamic Web Project. Select "6.0" for the Dynamic web module version.

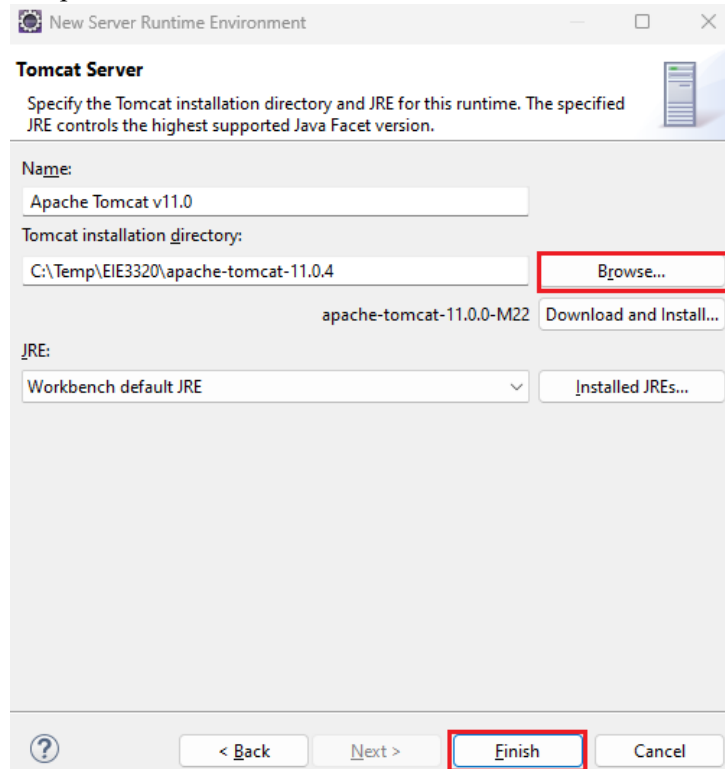


19. Click 'New Runtime...' to select the target runtime for first-time users."

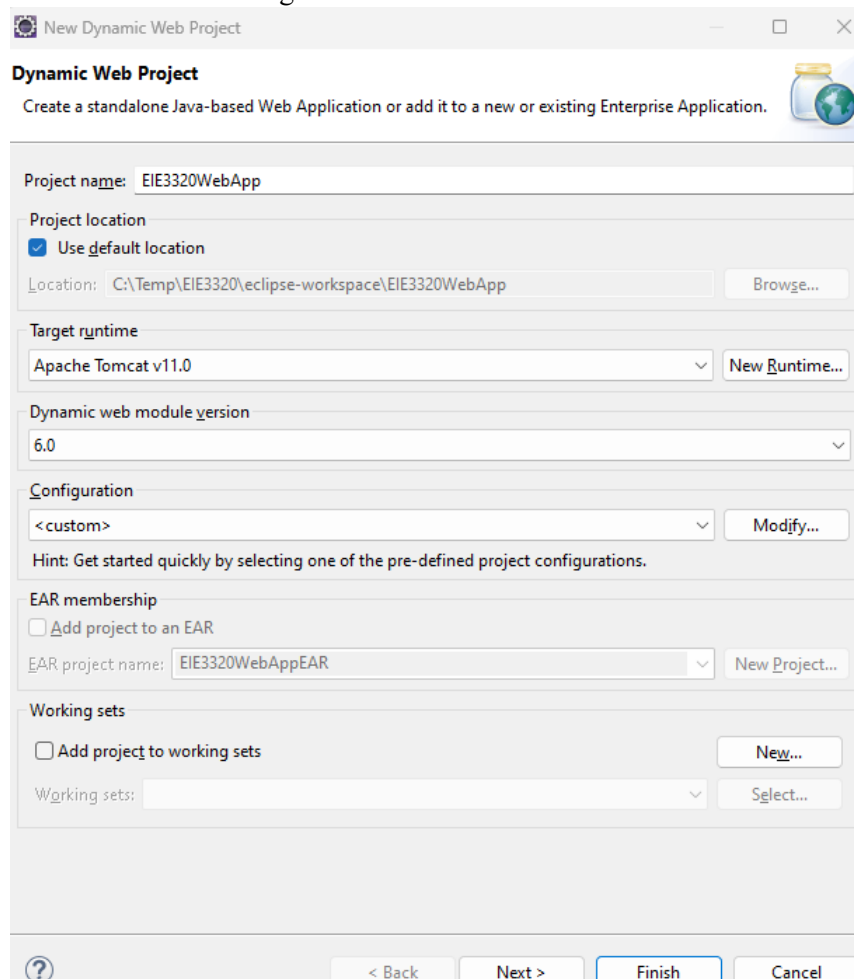
20. Select Apache Tomcat v11.0 and check "Create a new local server" for first-time user. Then, click Next.



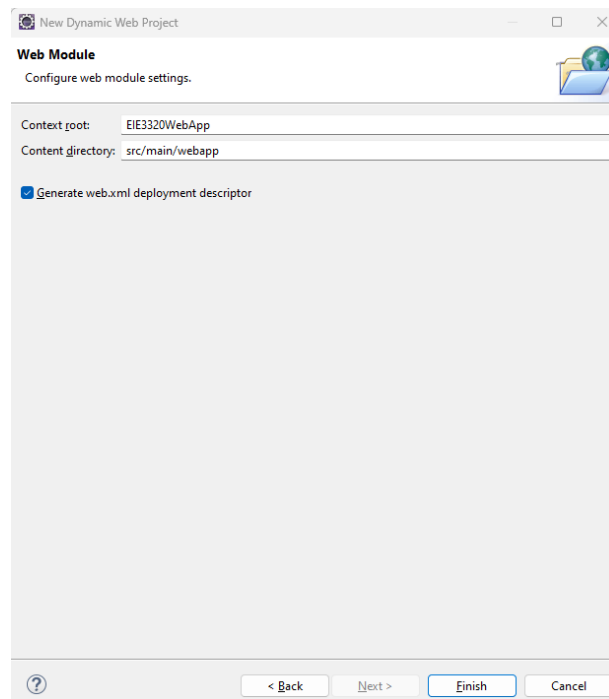
21. Click "Browse..." and select the uncompressed apache-tomcat folder. The next level of the folder path should be C:\Temp\EIE3320\apache-tomcat-11.0.4\bin. Then, click "Finish".



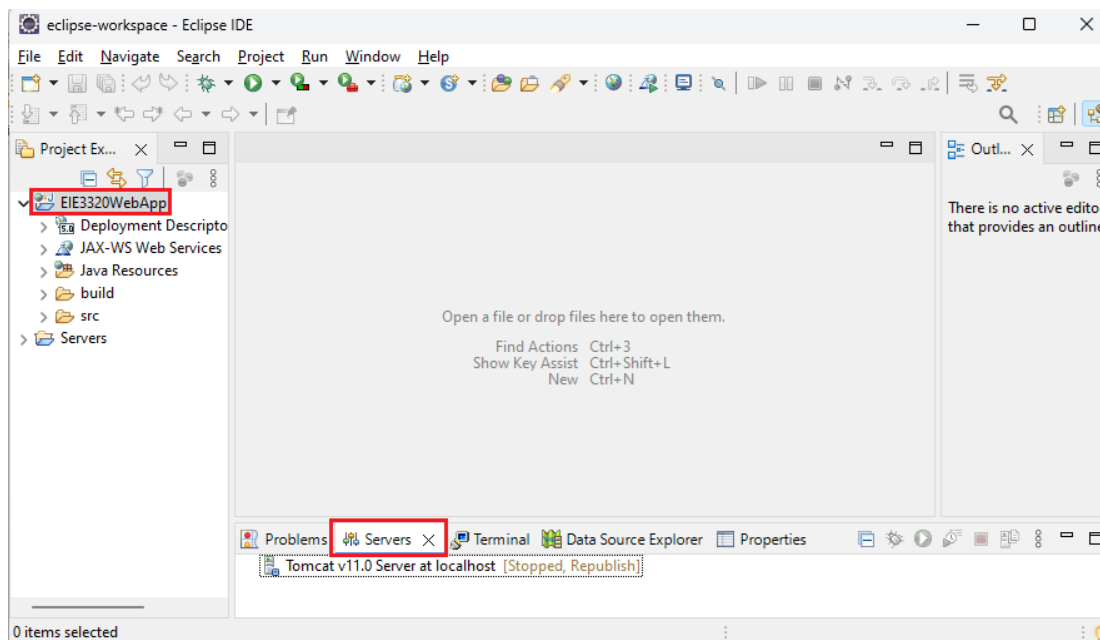
22. You will see a screen where Apache Tomcat v11.0 and Dynamic Web Module 6.0 should be selected. Then, press "Next >" twice for the following screen.



23. Check 'Generate web.xml deployment descriptor' and click 'Finish'



24. You can either click on "Servers" tab to check the server status, or right-click on the project name to create HTML, CSS, JavaScript, JSP, Class, or Servlet files.

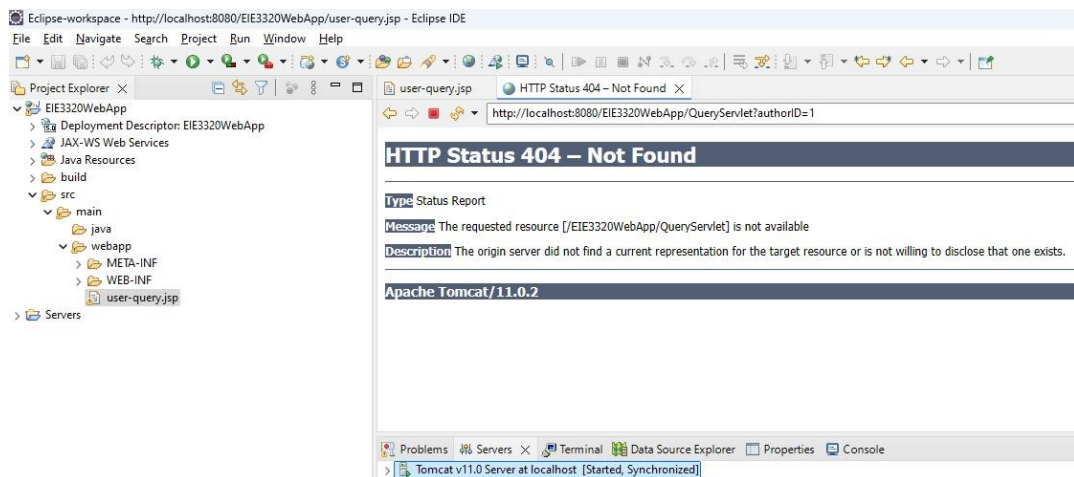
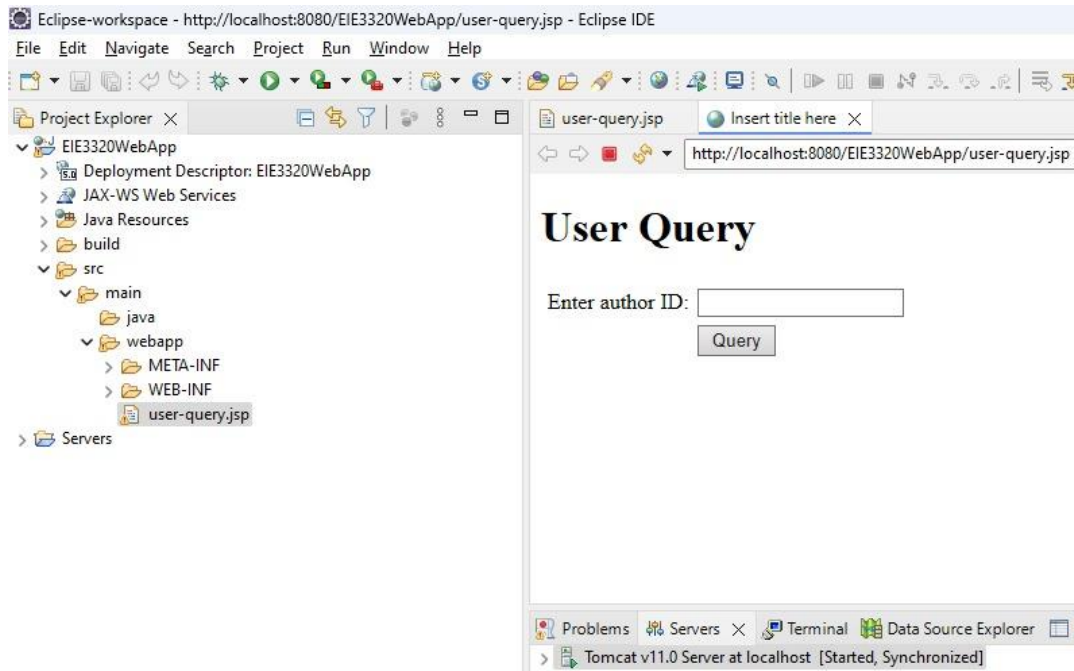


Section 2 – Homework 7

1. Create a JSP file called “user-query.jsp”. Right click the project title “EIE3320WebApp” in the Project Explorer on the left and select New -> JSP File. Add the following code to the <body></body> element of user-query.jsp. Save the file. This JSP file will send the user request to QueryServlet.class that runs on the server with the parameter authorID. When the server sends back a response, the JSP file retrieves the error object via the session attribute “error”.

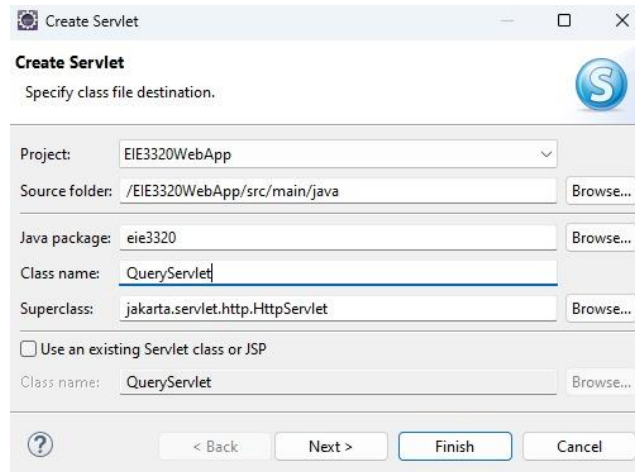
```
<h1>User Query</h1>
<%
    String error_message = "";
    Object error = request.getAttribute("error");    if
(error != null) error_message = error.toString();
%>
<form action="QueryServlet">
<table cellpadding="4">
<tr>
    <td>Enter author ID:</td>
    <td><input name="authorID" type="text" size="20"></td>
    <td style="color: red"><%= error_message %></td>
</tr>
<tr>
    <td></td>
    <td><input type="submit" value="Query"></td>
    <td></td>
</tr>
</table>
</form>
```

2. On the top menu, choose Window -> Web Browser, select “0 Internal Web Browser”. Execute “user-query.jsp” by right clicking the file in the Project Explorer (You may see the file by expanding “EIE3320WebApp” and then “WebContent”) and select Run As->Run on Server.
3. Choose the Tomcat v11.0 Server and click “Finish”. Follow the on screen instruction to start Tomcat automatically.
4. Input “1” to the edit box and press “Query”. You should see the following screen snapshots. Because you do not have “QueryServlet” yet, you will receive Error code 404.



Developing Servlets in Eclipse

5. Create a new servlet called “QueryServlet”. Right click the project “EIE3320WebAPP” ->New->Servlet. Use “eie3320” as package name and type QueryServlet as the Class name. If Eclipse cannot recognize servlet-related classes (e.g., HttpServlet), you may have done something wrong in earlier steps. In such case, you need to tell Eclipse where to find the file “servlet-api.jar”. To do that, right-click the project EIE3320WebApp, select Build Path, then Configure Build Path. In the Libraries tab, press the button “Add External JAR”, and then browse to “servlet-api.jar”. It should be in the folder <Tomcat Folder>/lib.

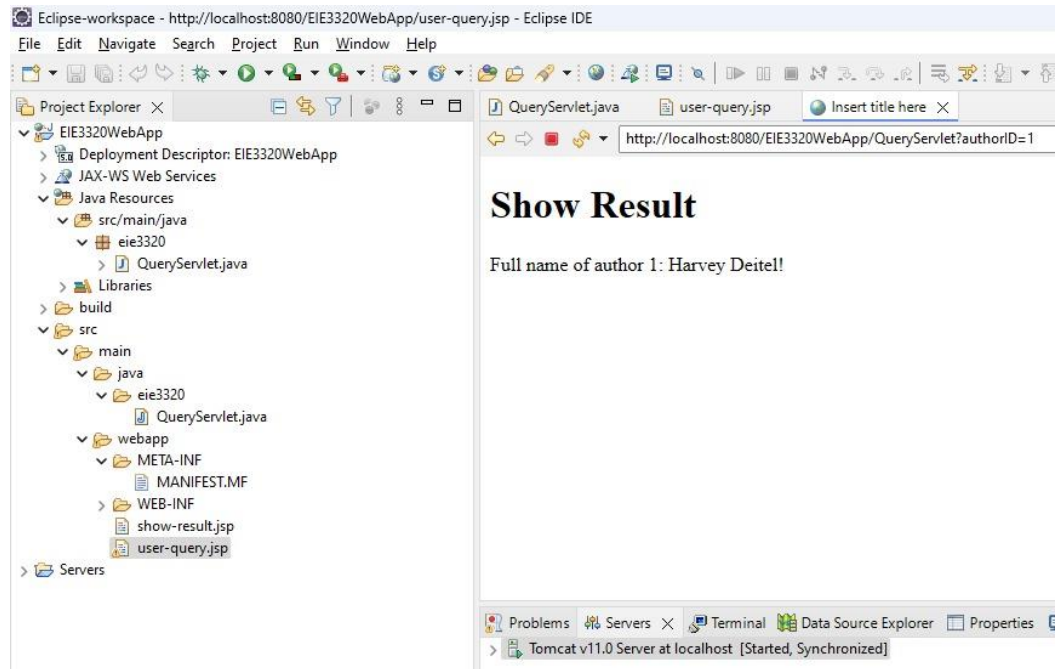


6. Execute user-query.jsp again and see what happen, remind that every changes in JSP and Servlet, you need to restart the server so as to reload the program. Explain your observation.

7. Create another JSP file called “show-result.jsp”. Add the following code to <body></body>

```
<h1>Show Result</h1>
<%
    String person = "?";
    String id = "?";
    Object authorID = request.getAttribute("authorID");
    if(authorID !=null)    id = authorID.toString();
    Object fullname = request.getAttribute("fullname");
    if(fullname !=null)    person =
fullname.toString();
%>
Full name of author <%=id%>:<%=person%> !
```

8. Modify “QueryServlet.java” until you can see the following screen snapshots. Replace “Harvey Deitel” by your name. Hints: You may need to set some attributes in the request scope of your servlet and use the classes Servlet Context and Request Dispatcher to forward the control to “show-result.jsp”. To understand the difference between attributes in request scope and session scope, see <https://stackoverflow.com/questions/4640721/servlets-setattribute-in-httpServletRequest-vs-setattributein-httpsession>



Programming

9. **(Complutory)** Create an HTML file called ATM.html and put the following code into the main body.

```
<form action="debit" method="POST">

  <p><strong>Please select the amount that you want to withdraw from your
account?</strong></p>

  <dl>

    <dd><input type="radio" name="Amount" value="1">1 </dd>

    <dd><input type="radio" name="Amount" value="10">10</dd>

    <dd><input type="radio" checked name="Amount" value="100">100</dd>

    <dd><input type="radio" name="Amount" value="1000">1000</dd>

  </dl>

  <p><input type="submit" value="Submit"> </p>
</form>

<form method="POST" action="credit">

  <p><strong>Please select the amount that you want to deposit in your
account?</strong></p>

  <dl>

    <dd><input type="radio" name="Amount" value="1">1 </dd>

    <dd><input type="radio" name="Amount" value="10">10</dd>

    <dd><input type="radio" checked name="Amount" value="100">100</dd>

    <dd><input type="radio" name="Amount" value="1000">1000</dd>

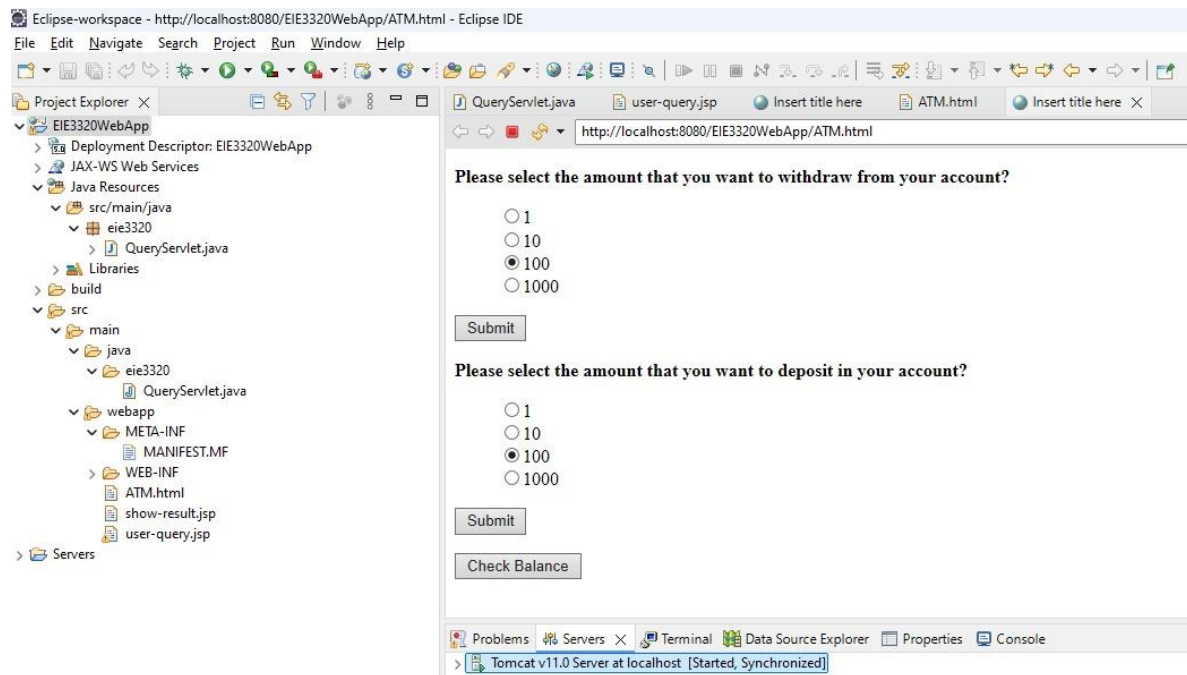
  </dl>

  <p><input type="submit" value="Submit" name="B1"></p>
</form>

<form method="GET" action="checkBalance">

  <p><input type="submit" value="Check Balance" name="B1"></p> </form>
```

Execute the HTML file by right clicking it and select Run As->Run on Server. You should see the following:



10. **(Complutory)** Create three servlets, namely CreditServlet.java, DebitServlet.java, and CheckBalanceServlet.java, so that a user accessing ATM.html can credit money, debit money, and check the balance of his/her account. Assume the initial balance is zero. Note that the servlets need to share a session-based object to keep track of the balance in the account.

1. After clicking “Check Balance”.

Your account balance is 0

[Click here to return to the main page](#)

2. After crediting \$100.

Your account balance is 100

[Click here to return to the main page](#)

3. After debiting \$1.

Your account balance is 99

[Click here to return to the main page](#)

Remember to compress the folder C:\Temp\EIE3320 and copy the compressed file to your flash memory before leaving the lab.

Submission for Assignment

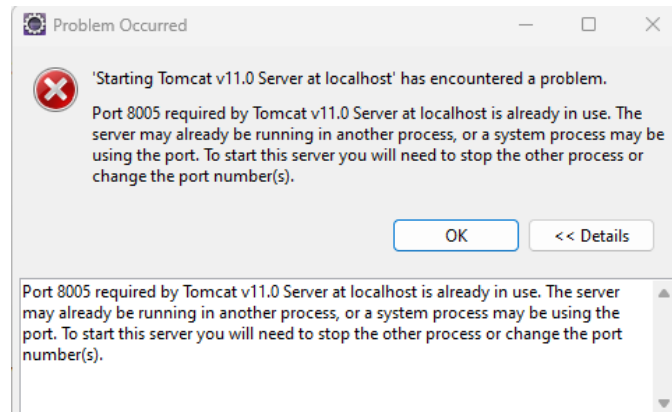
You should compress the Web application folder (including all source codes) under Workspace into Zip / 7z format and upload the zip file to Blackboard. DO NOT upload Tomcat server, SQL server, SDK development kit and other non-related programs. Otherwise, your mark WILL BE DEDUCTED.

Technical Tricks:

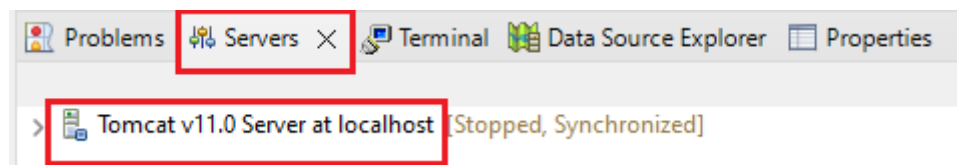
If you encounter “HTTP Status 404 Error”, chances are that Eclipse fails to deploy your webapps to the sever instance folder. In this situation, check whether your .jsp files and .html files can be found in the folder eclipse-workspace/.metadata/.plugins/org.eclipse.wst.server.core/tmp0/wtpwebapps/EIE3320WebApp/. To fix the problem, re-save your .jsp, .java, and .html files and re-run the webapps until Eclipse can deploy the tomcat instance to this folder.

APPENDIX 1 - Trouble shooting in EEE laboratories

1. How to change port



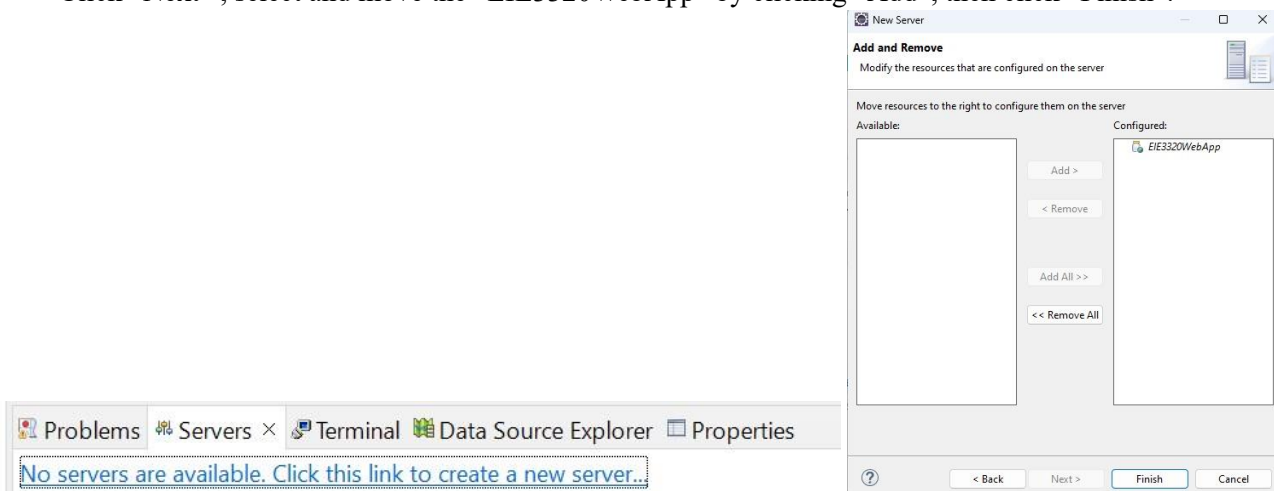
In EEE laboratory computers, the port 8005 is used by other application. Please change the port to 8006. Double click or Right click the Tomcat server created and select Open, the Overview of server will be shown.



Ports	
Modify the server ports.	
Port Name	Port Number
Tomcat admin port	8006
HTTP/1.1	8080

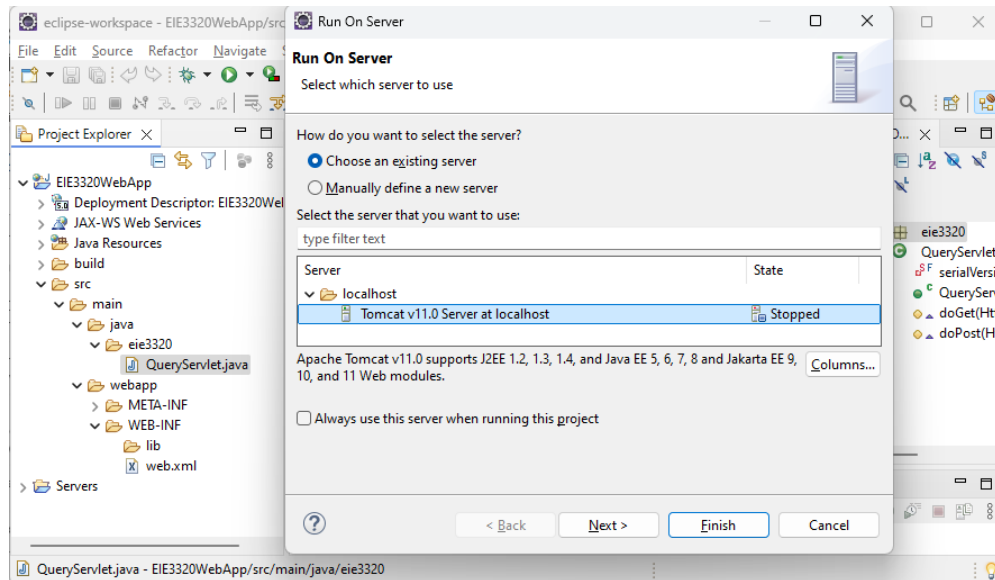
2. How to create web server

Create a new server by clicking “Server” on the bottom panel, Click “No servers are available. Click this link to create a new server...”. Under the server type, select “Tomcat v11.0 Server” in “Apache” folder. Click “Next” , select and move the “EIE3320WebApp” by clicking “Add”, then click “Finish”.



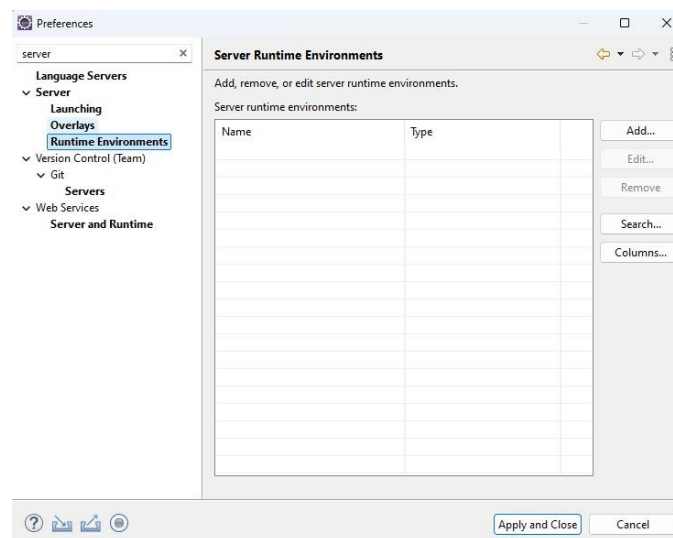
3. How to run file on web server

To run the servlet, right-click on the file and select Tomcat v11.0 Server in the "Run On Server" window.



4. How to setup and modify Runtime Environments

Setup Web Application Server. On the top menu, select Window->Preferences. In the left panel, expand “Server” then select “Run Time Environment”. Press “Add” and follow the instruction to select Apache Tomcat 11.0. Select C:\Temp\EIE3320\apache-tomcat-11.0.4 as the installation directory of Tomcat.



5. How to Inspect the Database Table

Install MySQL Workbench from <https://dev.mysql.com/downloads/workbench>. Ensure that you do not select the option to install the MySQL server.

MySQL Workbench

Local instance MySQL84 - Wamin.x Local instance MySQL84 - Wamin.x Local ins

File Edit View Query Database Server Tools Scripting Help

SQL

Navigator

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Reports
- Performance Schema Setup

Query 1 Administration - Data Export new_sc

```
1 • Use Books;  
2 • Select * from authors;  
3
```

Result Grid

	AuthorID	FirstName	LastName
▶	1	Harvey	Deitel
	2	Paul	Deitel
	3	Andrew	Goldberg
	4	David	Choffnes
*	NULL	NULL	NULL

APPENDIX 2 – MySQL commands

Connect mysql server

```
C:\Program Files\MySQL\MySQL Server 8.4\bin>mysql -u root -padmin_mysql
mysql: [Warning] Using a password on the command line interface can be insecure.
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 18
Server version: 8.4.1 MySQL Community Server - GPL

Copyright (c) 2000, 2024, Oracle and/or its affiliates.

Oracle is a registered trademark of Oracle Corporation and/or its
affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> |
```

cd "C:\Program Files\MySQL\MySQL Server 8.4\bin"

mysql -u root -padmin_mysql

Delete user “guest” if exists

```
mysql> use mysql;
Database changed
mysql> drop user if exists 'guest'@'localhost';
Query OK, 0 rows affected (0.00 sec)
```

use mysql;

drop user if exists 'guest'@'localhost';

Add user “guest”

```
mysql> create user 'guest'@'localhost' identified by 'guest';
Query OK, 0 rows affected (0.00 sec)

mysql> grant select, insert, update, delete, create, drop, references, execute on *.* to 'guest'@'localhost';
Query OK, 0 rows affected (0.00 sec)
```

create user 'guest'@'localhost' identified by 'guest';

*grant select, insert, update, delete, create, drop, references, execute on *.* to 'guest'@'localhost';*

Execute SQL Script to create tables and import data

```
mysql> source C:/Temp/EIE3320/MySQL/Books.sql;
```

source C:/Temp/EIE3320/MySQL/Books.sql;

Verification

```
mysql> show databases;
+-----+
| Database |
+-----+
| books    |
| information_schema |
| mysql    |
| performance_schema |
| sys      |
+-----+
5 rows in set (0.00 sec)

mysql> use books;
Database changed
mysql> show tables;
+-----+
| Tables_in_books |
+-----+
| authorisbn      |
| authors         |
| bookinfo        |
+-----+
3 rows in set (0.00 sec)

mysql> select * from authors;
+-----+-----+-----+
| AuthorID | FirstName | LastName |
+-----+-----+-----+
| 1        | Harvey   | Deitel   |
| 2        | Paul     | Deitel   |
| 3        | Andrew   | Goldberg |
| 4        | David    | Choffnes |
+-----+-----+-----+
4 rows in set (0.00 sec)
```

Show databases;

Use books;

Show tables;

*Select * from authors;*

```
mysql> update authors set FirstName="Test1", LastName="Test2" where AuthorID=1;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> select * from authors;
+-----+-----+-----+
| AuthorID | FirstName | LastName |
+-----+-----+-----+
| 1 | Test1 | Test2 |
| 2 | Paul | Deitel |
| 3 | Andrew | Goldberg |
| 4 | David | Choffnes |
+-----+-----+-----+
4 rows in set (0.00 sec)
```

update authors set FirstName="Test1", LastName="Test2" where AuthorID=1;

The screenshot shows the Eclipse IDE interface. On the left, the Project Explorer displays the structure of the 'EIE3320WebApp' project. The project contains a 'src/main/java' directory with a package 'eie3320' containing 'QueryServlet.java'. There are also 'Libraries', 'build', and 'src' directories. The 'src/main/webapp' directory contains 'META-INF' (with 'MANIFEST.MF'), 'WEB-INF' (with 'lib' and 'show-result.jsp'), and 'user-query.jsp'. The 'Servers' view shows 'Tomcat v11.0 Server at localhost-conf' with configuration files like 'catalina.properties', 'context.xml', 'server.xml', 'tomcat-users.xml', and 'web.xml'.

On the right, the browser window shows the URL 'http://localhost:8080/EIE3320Web'. The page content displays 'Show Result' in a large, bold, serif font, followed by the text 'Full name of author 1:Test1 Test2 !'.

APPENDIX 3 – Common SQL Statements

1. SELECT Statement

- **Purpose:** Retrieves data from a database table.
- **Syntax:** `SELECT column1, column2 FROM table_name;`
- **Example:**
`SELECT CustomerName, City FROM Customers;`

2. SELECT with Different Columns from Two Tables

- **Purpose:** Retrieves specific columns from two tables.
- **Syntax:**
`SELECT table1.column1, table2.column2
FROM table1, table2
WHERE table1.common_column = table2.common_column;`
- **Example:**
`SELECT orders.order_id, customers.customer_name
FROM orders, customers
WHERE orders.customer_id = customers.customer_id;`

3. INSERT INTO Statement

- **Purpose:** Inserts new records into a table.
- **Syntax:** `INSERT INTO table_name (column1, column2) VALUES (value1, value2);`
- **Example:**
`INSERT INTO Customers (CustomerName, City) VALUES ('New Customer', 'New York');`

4. UPDATE Statement

- **Purpose:** Modifies existing records in a table.
- **Syntax:** `UPDATE table_name SET column1 = value1 WHERE condition;`
- **Example:**
`UPDATE Customers SET City = 'Paris' WHERE CustomerName = 'Alfreds Futterkiste';`

5. DELETE Statement

- **Purpose:** Deletes records from a table.
- **Syntax:** `DELETE FROM table_name WHERE condition;`
- **Example:**
`DELETE FROM Customers WHERE CustomerName = 'New Customer';`

6. CREATE TABLE Statement

- **Purpose:** Creates a new table in a database.
- **Syntax:** `CREATE TABLE table_name (column1 datatype, column2 datatype);`
- **Example:**

-- Drop the table if it exists

```
DROP TABLE IF EXISTS Employees;
```

```
CREATE TABLE Employees (  
    EmployeeID int,  
    Name varchar(255),  
    Department varchar(255)  
);
```


APPENDIX 4 – Special settings are required to connect to the MySQL database pre-installed as a server service in CF502/CF504

(Note: MySQL can be installed using either MySQL Installer or MySQL Workbench)

Connect mysql server

cd "C:\Program Files\MySQL\MySQL Server 8.4\bin"

mysql -u root -padmin _mysql

(Replace admin _mysql with actual password)

```
C:\Program Files\MySQL\MySQL Server 8.4\bin>mysql -u root -padmin_mysql
mysql: [Warning] Using a password on the command line interface can be insecure.
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 18
Server version: 8.4.1 MySQL Community Server - GPL

Copyright (c) 2000, 2024, Oracle and/or its affiliates.

Oracle is a registered trademark of Oracle Corporation and/or its
affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> |
```

Delete user “guest” if exists

use mysql;

drop user if exists 'guest'@'localhost';

```
mysql> use mysql;
Database changed
mysql> drop user if exists 'guest'@'localhost';
Query OK, 0 rows affected (0.00 sec)
```

Add user “guest”

create user 'guest'@'localhost' identified by 'guest';

*grant select, insert, update, delete, create, drop, references, execute on *.* to 'guest'@'localhost';*

```
mysql> create user 'guest'@'localhost' identified by 'guest';
Query OK, 0 rows affected (0.00 sec)

mysql> grant select, insert, update, delete, create, drop, references, execute on *.* to 'guest'@'localhost';
Query OK, 0 rows affected (0.00 sec)
```

Execute SQL Script to create tables and import data

source C:/Temp/EIE3320/MySQL/Books.sql;

```
mysql> source C:/Temp/EIE3320/MySQL/Books.sql;
```

Verification

Show databases;

Use books;

Show tables;

*Select * from authors;*

```
mysql> show databases;
+-----+
| Database |
+-----+
| books    |
| information_schema |
| mysql    |
| performance_schema |
| sys      |
+-----+
5 rows in set (0.00 sec)
```

```
mysql> use books;
Database changed
mysql> show tables;
+-----+
| Tables_in_books |
+-----+
| authorisbn      |
| authors         |
| bookinfo        |
+-----+
3 rows in set (0.00 sec)
```

```
mysql> select * from authors;
+-----+-----+-----+
| AuthorID | FirstName | LastName |
+-----+-----+-----+
| 1        | Harvey   | Deitel   |
| 2        | Paul     | Deitel   |
| 3        | Andrew   | Goldberg |
| 4        | David    | Choffnes |
+-----+-----+-----+
4 rows in set (0.00 sec)
```

APPENDIX 5 – Special setting to change MySQL port number

(If you wish to host multiple MySQL servers on different ports or if the default port is used by another application.)

After connecting to the MySQL server in the command prompt, type:

```
mysqld --initialize-insecure
```

```
mysqld -port 3320
```

```
P:\>netstat -ano | find /i "3320"
TCP    0.0.0.0:3320          0.0.0.0:0          LISTENING        12208
TCP    [::]:3320          [::]:0             LISTENING        12208
```

```
mysql -h localhost -u root -port 3320
```

```
mysqladmin -h localhost -u root shutdown --port 3320
```

```
mysql -h localhost -u guest -pguest --port 3320
```

Change of database connection string:

```
String url = "jdbc:mysql://localhost:3320/books";
```